

AI ASSISTED CODING ASS-7.3

2303A51479

B-27

Task 1: Fixing Syntax Errors

Scenario: You are reviewing a Python program where a basic function definition contains a syntax error.

Python function with Syntax Error:

```
def add(a, b)  
    return a + b
```

```
/HARSHAVARDHAN/Desktop/AI Assitant Coding/addition.py"  
File "c:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding\addition.py", line 1  
def add(a, b)  
    ^  
SyntaxError: expected ':'
```

Corrected Python function:

```
def add(a, b):  
    return a + b
```

Explanation of the fix:

Error: the function header was missing the required ":" which causes a SyntaxError because Python's parser expects a colon to start the function suite.

Fix: add ":" after def and ensure the body is indented.

Task 2: Debugging Logic Errors in Loops

Scenario: You are debugging a loop that runs infinitely due to a logical mistake.

Given Code:

```
def count_down(n):  
    while n >= 0:  
        print(n)  
        n += 1 #Should be n -= 1 to count down
```

AI Corrected Code:

```
def count_down(n):  
    while n >= 0:  
        print(n)  
        n -= 1
```

Explanation:

Error: The loop uses `n += 1` which increments `n`, causing an infinite loop instead of counting down. The condition while `n >= 0` will always be true.

Fix: Change `n += 1` to `n -= 1` to decrement `n` each iteration.

The `count_down` function should decrease `n` by 1 each iteration until it reaches below 0.

Using `+=` does the opposite, making the loop infinite since `n` keeps growing and will always be `>= 0`.

Task 3: Handling Runtime Errors (Division by Zero)

Scenario: A Python function crashes during execution due to a division by zero error.

Given Code:

```
def divide(a, b):  
    return a / b  
  
print(divide(10, 0)) # This will raise a ZeroDivisionError
```

```
File "c:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding\main.py", line 12, in <module>  
    print(divide(10, 0)) # This will raise a ZeroDivisionError  
~~~~~
```

```
File "c:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding\main.py", line 10, in divide  
    return a / b  
~~~~~
```

```
ZeroDivisionError: division by zero
```

AI Corrected Code:

```
def divide(a, b):  
    if b == 0:  
        return "Error: Cannot divide by zero"  
    return a / b  
  
print(divide(10, 0)) # Returns error message instead of crashing
```

Explanation:

Error: Division by zero causes a `ZeroDivisionError` when `b = 0`. The code attempts to divide 10 by 0, which is mathematically undefined.

Fix: Add error handling using a try-except block to catch the exception.

Before performing division, check if `b` is 0. If it is, return an error message instead of attempting the division. This prevents the `ZeroDivisionError` from crashing your program. Alternatively, you can use try-except for more complex error handling.

Task 4: Debugging Class Definition Errors

Scenario: You are given a faulty Python class where the constructor is incorrectly defined.

Given Code:

```
class Rectangle:
    def __init__(length, width):
        self.length = length
        self.width = width
```

AI Corrected Code:

```
class Rectangle:
    def __init__(self, length, width):
        self.length = length
        self.width = width
```

Explanation:

Error: The `__init__` method is missing `self` as the first parameter. This causes a `TypeError` when creating a `Rectangle` instance.

Fix: Add `self` as the first parameter.

All instance methods in Python classes must have `self` as the first parameter. `self` refers to the instance being created and allows you to set instance attributes like `self.length` and `self.width`. Without it, the method doesn't know which object's attributes to modify.

Task 5: Resolving Index Errors in Lists

Scenario: A program crashes when accessing an invalid index in a list.

Given Code:

```
numbers = [1, 2, 3]
print(numbers[5]) # This will raise an IndexError
```

```
File "c:\Users\HARSHAVARDHAN\Desktop\AI Assitant Coding\main.py", line 24, in <module>
    print(numbers[5]) # This will raise an IndexError
          ~~~~~^
IndexError: list index out of range
```

AI Corrected Code:

```
numbers = [1, 2, 3]
#Access valid index
print(numbers[2]) # Prints 3
```

Explanation:

Error: IndexError - attempting to access index 5 in a list that only has 3 elements (indices 0, 1, 2). Array indices are out of bounds.

Fix: Check the list length or use valid index values.

Lists in Python are zero-indexed, so a list with 3 elements has valid indices 0, 1, and 2. Accessing index 5 doesn't exist, causing an IndexError. Always verify the index is within bounds before accessing, or use error handling with try-except.